Software Testing & QA Automation

Complete Program Syllabus

Program Overview

Modules at a Glance

Module 1: Foundations of Software Testing & QA

1.1 Software Testing Fundamentals

1.2 The Software Testing Life Cycle (STLC)

1.3 Test Design Techniques

1.4 Defect Management

1.5 Test Documentation

Module 2: Manual Testing Mastery

2.1 Functional Testing

2.2 Non-Functional Testing

2.3 Specialized Testing Types

2.4 Agile & Exploratory Testing

2.5 Test Management Tools

Module 3: Test Automation Foundations & Selenium

3.1 Automation Testing Fundamentals

3.2 Programming for Automation (Java/Python)

3.3 Selenium WebDriver

3.4 Locators & Synchronization

3.5 Design Patterns & Frameworks

3.6 TestNG / JUnit & Reporting

Shrinovatech

Module 4: Modern End-to-End Testing with Playwright

- 4.1 Playwright Fundamentals
- 4.2 Advanced Playwright Features
- 4.3 Cross-Browser & Visual Testing
- 4.4 Playwright Test Runner &

Reporting Module 5: API

Testing & Service Automation

- 5.1 API Testing Fundamentals
- 5.2 Postman for Manual & Automated API Testing
- 5.3 API Automation with Code (REST Assured / Requests)
- 5.4 Contract & Performance Testing Basics
- 5.5 Database & Backend Validation

Module 6: CI/CD, DevOps & Test Management

- 6.1 Version Control for QA
- 6.2 CI/CD Pipelines for Automated Testing
- 6.3 Containerization & Test Environments
- 6.4 Test Reporting & Dashboards
- 6.5 Performance & Load Testing
- 6.6 QA Leadership & Best Practices

Shrinovatech

Program Overview

This program takes learners from the foundations of manual software testing through industry-standard automation with Selenium and Playwright, API testing with Postman and
code-based frameworks, and finally into CI/CD-integrated, production-grade QA practice. Each module builds directly on the last and ends with a hands-on project.

Modules at a Glance

No	Module	Focus	Duration
1	Foundations of Software Testing &	QCAore testing concepts & STLC	1–2 weeks
2	Manual Testing Mastery	Functional & non-functional testing	2–3 weeks
3	Test Automation Foundations & Selenium	Web Driver, frame works, Selenium	3–4 weeks
4	Modern E2E Testing with Playwright	Playwright automation & visual testing	2–3 weeks
5	API Testing & Service Automation	Postman, REST Assured, contract testing	2–3 weeks
6	CI/CD, DevOps & Test Management	Pipelines, Docker, reporting, performance	2–3 weeks

Shinovatech

Module 1: Foundations of Software Testing & QA

Goal: Build a solid grounding in QA principles, terminology, and the testing life cycle that every later module depends on.

1.1 Software Testing Fundamentals

- What is software testing? Verification vs. validation
- Principles of testing; testing vs. debugging
- Types of testing: manual vs. automated, static vs. dynamic
- QA vs. QC vs. Testing — roles and responsibilities
- SDLC models and where testing fits (Waterfall, Agile, V-Model)

1.2 The Software Testing Life Cycle (STLC)

- Requirement analysis and test planning
- Test case design and review
- Test environment setup
- Test execution and closure
- Entry and exit criteria

1.3 Test Design Techniques

- Black-box techniques: equivalence partitioning, boundary value analysis
- Decision table testing, state transition testing
- White-box basics: statement, branch, and path coverage
- Writing effective, traceable test cases

1.4 Defect Management

- Defect life cycle and severity vs. priority
- Writing high-quality bug reports
- Defect tracking tools (Jira, Azure DevOps, Bugzilla)
- Root cause analysis basics

1.5 Test Documentation

- Test plans and test strategy documents
- Test cases, test scripts, and checklists
- Requirement Traceability Matrix (RTM)

Shrinovatech

- Test summary and closure reports

Module Project:	Write a complete test plan and a set of traceable manual test cases (with an RTM) for a given web application requirement document.

Module 2: Manual Testing Mastery

Goal: Develop deep, practical skill in manual testing across functional, non-functional, and specialized testing types used in real projects.

2.1 Functional Testing

- Smoke, sanity, and regression testing
- Unit, integration, and system testing (QA perspective)
- User Acceptance Testing (UAT)
- Cross-browser and cross-device functional checks

2.2 Non-Functional Testing

- Usability and accessibility testing basics
- Compatibility testing
- Introduction to performance, load, and stress testing
- Security testing fundamentals (OWASP Top 10 awareness)

2.3 Specialized Testing Types

- API testing overview (manual)
- Database testing and data validation
- Mobile app testing considerations
- Localization and internationalization testing

2.4 Agile & Exploratory Testing

- Testing in Scrum and Kanban teams
- Exploratory testing charters and session-based testing
- Risk-based testing and test prioritization
- Collaborating with developers and product owners

Shrinovatech

2.5 Test Management Tools

- Jira, TestRail, Zephyr for test case management
- Organizing test suites, cycles, and runs
- Reporting test coverage and execution status

Module Project :	Perform full manual QA on a sample e-commerce web app: design and execute a functional and exploratory test cycle, log defects in Jira, and produce a test summary report.

Module 3: Test Automation Foundations & Selenium

Goal: Learn the fundamentals of test automation and become proficient with Selenium WebDriver, the industry-standard tool for web UI automation.

3.1 Automation Testing Fundamentals

- When to automate vs. when to keep tests manual
- Automation pyramid: unit, API, UI tests
- Choosing an automation tool and framework
- ROI of automation and maintenance considerations

3.2 Programming for Automation (Java/Python)

- Core syntax, OOP concepts, collections
- Exception handling and file I/O
- Build tools: Maven/Gradle (Java) or pip/venv (Python)
- IDE setup: IntelliJ / VS Code

3.3 Selenium WebDriver

- Selenium architecture and browser drivers
- Setting up WebDriver, launching and controlling browsers
- WebElement interactions: click, type, select, hover
- Handling alerts, frames, windows, and dropdowns

3.4 Locators & Synchronization

Shrinovatech

- Locator strategies: ID, CSS selectors, XPath
- Implicit vs. explicit waits, fluent waits
- Handling dynamic elements and flaky tests

3.5 Design Patterns & Frameworks

- Page Object Model (POM) and Page Factory
- Data-driven testing with external data sources
- Building a maintainable automation framework from scratch
- Keyword-driven and hybrid frameworks (conceptual)

3.6 TestNG / JUnit & Reporting

- Annotations, assertions, and test grouping
- Parallel execution and cross-browser test suites
- Extent Reports / Allure for automated texst reporting

Module Project :	Build a Page Object Model-based Selenium automation framework for a demo web application, including data-driven tests, TestNG suites, and an HTML test report.

Shrinotech

Module 4: Modern End-to-End Testing with Playwright

Goal: Master Playwright as a modern, fast, and reliable alternative to Selenium for end-to-end web automation.

4.1 Playwright Fundamentals

- Playwright architecture: browser contexts and auto-waiting
- Installation and project setup (JavaScript/TypeScript or Python)
- Locators, actions, and assertions
- Handling navigation, dialogs, and iframes

4.2 Advanced Playwright Features

- Network interception and request mocking
- Authentication state reuse across tests
- File uploads/downloads and browser storage
- Component and API testing within Playwright

4.3 Cross-Browser & Visual Testing

- Running tests across Chromium, Firefox, and WebKit
- Mobile emulation and responsive testing
- Visual regression testing with screenshot comparison

4.4 Playwright Test Runner & Reporting

- Test fixtures, hooks, and parallelization
- Trace viewer and debugging failed tests
- HTML reports and CI-friendly output formats

Module Project :	Convert a subset of the Module 3 Selenium suite into Playwright, add visual regression checks and network mocking, and run the suite across three browser engines.

chrinovatech

Module 5: API Testing & Service Automation

Goal: Gain hands-on expertise testing REST APIs both manually with Postman and programmatically within automation frameworks.

5.1 API Testing Fundamentals

- REST principles, HTTP methods, status codes
- Request/response structure, headers, and payloads
- Authentication: API keys, OAuth 2.0, JWT

5.2 Postman for Manual & Automated API Testing

- Building and organizing collections and environments
- Writing test scripts and assertions in Postman
- Chaining requests, variables, and pre-request scripts
- Newman for running Postman collections from the CLI

5.3 API Automation with Code (REST Assured / Requests)

- Building an API test framework in Java (REST Assured) or Python (Requests + pytest)
- Schema validation (JSON Schema)
- Data-driven API tests and reusable request builders

5.4 Contract & Performance Testing Basics

- Introduction to contract testing (Pact, conceptual)
- Mock servers and service virtualization
- Basic load testing of APIs (k6 or JMeter, conceptual)

5.5 Database & Backend Validation

- Writing SQL queries for backend data validation
- Validating API responses against database state
- End-to-end data integrity checks

Module Project :	Build a Postman collection with full test scripts for a public REST API, then reproduce the same coverage as a code-based automated test suite that runs via Newman/CLI.

Shrinovatech

Module 6: CI/CD, DevOps & Test Management

Goal: Integrate automated tests into real delivery pipelines and manage QA at team and organizational scale.

6.1 Version Control for QA

- Git fundamentals: branches, commits, pull requests
- Collaborating on shared automation repositories
- Code review practices for test code

6.2 CI/CD Pipelines for Automated Testing

- Jenkins pipelines for scheduled and triggered test runs
- GitHub Actions / GitLab CI for automated test execution
- Integrating Selenium/Playwright suites into pipelines

6.3 Containerization & Test Environments

- Docker basics for running tests in containers
- Selenium Grid and containerized browser farms
- Managing test data and environment configuration

6.4 Test Reporting & Dashboards

- Allure, ExtentReports, and CI-integrated dashboards
- Test result trends and flaky test tracking
- Communicating quality metrics to stakeholders

6.5 Performance & Load Testing

- JMeter or k6 fundamentals
- Designing load, stress, and soak test scenarios
- Analyzing performance test results

6.6 QA Leadership & Best Practices

- Building a test strategy for a new project
- Shift-left and shift-right testing practices

Shrinovatech

- Estimating and planning QA effort in Agile teams
- Career paths: SDET, QA Lead, Automation Architect

Module Project :	Set up a CI/CD pipeline (GitHub Actions or Jenkins) that runs your Selenium, Playwright, and API test suites in Docker on every commit, publishes an Allure report, and includes a basic JMeter/k6 performance check.

Capstone Project: Design and deliver a complete QA automation suite for a real (or sample) web application — combining manual test documentation, a Selenium or Playwright UI framework, a Postman/code-based API test suite, and a CI/CD pipeline with automted reporting and a basic performance test.

Shrinovatech